

RAJESH THULASI

Technical R&D Head



(Immediate Joiner)

CONTACT DETAILS

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EDUCATION

2006: Bachelor of Mechanical Engineering from Anna University, Chennai, Tamil Nadu

2002: Diploma, Mechanical Engineering from Government Polytechnic, Anantapur, Andhra Pradesh

1999: SSC from ZP High School, Nagari, Andhra Pradesh

CORE COMPETENCIES

- ➔ Automotive ECU Development
- ➔ Electromechanical Systems
- ➔ Embedded HW & EMI / EMC
- ➔ ISO 26262 Compliance
- ➔ ISO/SAE 21434 certified
- ➔ Cross-Functional Leadership
- ➔ Mechanical Design Optimization
- ➔ Thermal & Structural Simulation
- ➔ DFM/DFA & DFMEA
- ➔ Global Program Management
- ➔ Cost Benchmarking
- ➔ Supplier Development
- ➔ CATIA V5 / SolidWorks
- ➔ Talent Development & Team Scaling
- ➔ Agile Hardware Development
- ➔ APQP / PPAP / IATF 16949

PERSONAL DETAILS

Date of Birth: 11th September 1983
Languages Known: English, Tamil, Telugu & Hindi
Present Address: Bangalore

JOB OBJECTIVE

Seeking to lead transformative **R&D hardware programs** that shape the future of **automotive ECUs and sensor technologies**, with a focus on delivering robust, reliable, and cost-effective solutions. Dedicated to advancing **hardware design, simulations, and system integration**, while ensuring seamless collaboration with **software teams** to enable cohesive, **software-defined vehicle platforms**. Committed to **building high-performing teams**, optimizing resources, and **driving innovation** pipelines that **accelerate product realization**. Focused on enabling **intelligent, connected, and sustainable mobility solutions** through next-generation hardware innovation.

PROFILE SUMMARY

- ➔ **Dynamic R&D Manager with 20+ years of expertise** in full-lifecycle product development of automotive ECUs and sensor systems, blending mechanical engineering acumen with deep material science, structural/thermal simulations, and platform scalability.
- ➔ **Spearheading hardware development** at Continental Automotive Components India while leading high-performance teams to meet stringent automotive benchmarks.
- ➔ **Success at leading cross-functional coordination** across four specialized teams - Mechanical Design, Simulations, Benchmarking, and Series Activities, while seamlessly collaborating with other key disciplines such as PCB Design, Hardware, and Systems Engineering to drive product development excellence and ensure alignment with project milestones.
- ➔ **Proven excellence in mechanical design optimization, global program execution**, regulatory compliance and engineering team scalability.
- ➔ **Successfully launched complex ECU programs under compressed timelines** and initiated strategic cost optimization initiatives projected to deliver annual savings, reinforcing value engineering and fiscal discipline.
- ➔ **Adept in high-end simulation tools and technologies**, combined with deep material science knowledge to deliver robust, reliable mechanical solutions that align with demanding automotive performance standards.
- ➔ **Recognized for fostering an innovation-driven ecosystem through mentoring, collaborative leadership**, and continuous improvement, enhancing cross-functional synergies, talent retention, and project velocity.
- ➔ Driving seamless hardware-software collaboration as a project leader, ensuring synchronized development between **embedded software, hardware, system engineering, functional safety, cybersecurity, quality, advanced purchasing, and plant manufacturing** teams to accelerate **ECU innovations** and enable software-defined vehicle strategies.

AWARDS & ACCOLADES

- ➔ Spearheaded a cross-functional cost benchmarking initiative, successfully implementing 25+ cost-optimization solutions across mechanical components, resulting in significant efficiency gains and material cost reductions.
- ➔ Engineered an 18% reduction in ECU size and eliminated the need for active cooling by designing a fin-optimized aluminum enclosure housing using advanced thermal simulation techniques, enhancing product reliability and manufacturability.
- ➔ Built and led a high-performing 55+ member cross-functional R&D team, driving the successful launch of the first ECU within just 14 months, ahead of industry benchmarks.
- ➔ Strategically transitioned 40% of mechanical development activities to India from European teams, realizing annualized cost savings exceeding ₹1.2 Crore while maintaining global quality and performance standards.

CERTIFICATION	TRAINING
Certified Automotive Cybersecurity ISO/SAE 21434 practitioner from TUV SUD.	PMP Training completion Certification from Knowledge Wood.

WORK EXPERIENCE

Sep 2015 – Nov -2025: AUMOVIO (Continental Automotive Components) India, Bengaluru

Growth Path:

Technical Specialist → Senior Technical Specialist → Senior Architect → **Lead Architect: R&D Head Hardware Engineering**

KEY RESULT AREAS

- ➔ **Innovation Pipeline Management:** Driving a structured innovation pipeline by identifying emerging technologies, prioritizing high-impact projects, and fostering a culture of continuous improvement and IP generation through patents and publications.
- ➔ **Strategic Planning & Resource Management:** Developing and executing long-term R&D strategies while optimizing budgets, resource allocation, and talent development to accelerate innovation cycles and maximize ROI on technical investments; ensuring **system-level alignment of hardware, software, and quality roadmaps**.
- ➔ **Product Development:** Orchestrating the end-to-end hardware development lifecycle for advanced automotive ECUs and sensor modules — spanning concept ideation, virtual simulation, rapid prototyping, design validation, and ramp-to-production readiness, while embedding **ASPICE-aligned requirement traceability and quality planning across hardware, software, and systems engineering teams**.
- ➔ **Project Leadership:** Leading mechanical engineering (ME) project management for key systems such as the Damping & Air Spring Controller, Current Sensing Module (CSM) for high-voltage batteries, and Intelligent Battery Sensor (IBS) for low-voltage platforms — focusing on design scalability, manufacturability, and robust performance under stringent automotive constraints, with **cross-domain integration of embedded software, diagnostics, and system validation**.
- ➔ **Design Optimization:** Championing mechanical architecture optimization through application of DFMEA, DFM/DFA, tolerance analysis, and strategic material engineering — leveraging plastic injection molding, die casting, sheet metal forming, and precision aluminum machining to meet thermal, vibration, fatigue, and EMC compliance.
- ➔ **Simulation & Validation:** Driving early-phase design validation using advanced simulation tools such as ANSYS, Abaqus, and while applying GD&T and tolerance stack-up methodologies to enhance dimensional robustness and field reliability.
- ➔ **Domain Expertise:** Demonstrating subject matter expertise in sensor packaging, thermal interface design, electromagnetic shielding strategies, and rugged ECU enclosure development suited for harsh environments including engine bay, underbody, and passenger cabin zones.
- ➔ **Team Building:** Building and nurturing a global mechanical engineering talent ecosystem — hiring, mentoring, and upskilling teams to deliver technical depth, CAD/CAE proficiency (CATIA V5, SolidWorks), and high-impact collaborative output.
- ➔ **Cross-functional Integration:** Collaborating cross-functionally with embedded software, hardware, systems, and validation teams to ensure cohesive product realization and compliance with automotive standards including **ISO 26262 (Functional Safety)**, **ISO/SAE 21434 (Cybersecurity)**, **ASPICE**, and **IATF 16949 quality frameworks**.
- ➔ **Operational Excellence:** Cultivating a culture of continuous improvement and engineering excellence by defining clear project KPIs, instituting agile development frameworks, and embedding lean methodologies in design and development workflows; strengthening **software and system quality gates** for first time-right delivery.
- ➔ **Talent Management:** Leading end-to-end talent management including competency mapping, onboarding, structured learning paths, goal setting, KPI tracking, and performance appraisals to retain and grow high-caliber talent.
- ➔ **Cost Engineering:** Responding to RFIs/RFQs by conducting thorough cost-benefit analysis, mechanical design feasibility assessments, and material strategy definition in alignment with commercial and technical goals; balancing cost efficiency with **quality and compliance requirements in hardware–software co-development**.
- ➔ **Supplier Development:** Overseeing supplier engagement and component readiness — driving on-time delivery, quality assurance, and vendor capability development through proactive collaboration and rigorous technical governance, including **software vendor quality assessments and compliance audits**.
- ➔ **Cost Center Budget Ownership:** Managing multi-million R&D cost center budgets, driving resource optimization, CAPEX/OPEX planning, and aligning spend with innovation priorities and organizational growth objectives
- ➔ **Global Leadership & Change Management:** Building and leading high-performing global R&D teams across hardware, software, and systems domains; fostering innovation-driven culture, mentoring future leaders, and steering organizational change to meet evolving **software-defined vehicle and quality compliance demands**

PREVIOUS WORK EXPERIENCE

Duration	Designation	Company / Location	Client	Project
Nov 2014 – Aug 2015	Technical Lead	Alexander Dennis – United Kingdom (Deputation from Caresoft Global)		ADL Enviro 500 & ADL Long Bus
Mar 2012 – Oct 2014	Specialist	Satyam Venture Engineering Services – Hyderabad	Qoros Automotive, China	Qoros 4 Sedan & Hatchback
Dec 2010 – Oct 2011	Senior Design Engineer	Daimler India Commercial Vehicles – Chennai (via CADES Digitech)	—	Daimler 25T (6x2) Program
Apr 2009 – Nov 2010	Design Engineer	Ashok Leyland (Deputed from Innovative Solutions)		
Aug 2007 – Apr 2009	Design Engineer	Infotech Enterprises (now Cyient Ltd.) – Bengaluru	TEREX, USA & PRATT & WHITENEY	TEREX HDT
May 2005 – Aug 2007	CAD Engineer	Clyde Pumps SPX – Bengaluru (via Doux Technologies – Chennai)	—	OH2 Centrifugal Pump